Geographic Epistemic Asymmetries in LLM-Assisted Policy Research: A 15-Country, 6-Model, 4-Language Benchmark

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Abstract

Large language models (LLMs) are increasingly used by researchers worldwide to support applied work in public policy, yet the extent and mechanism of their geographic factual bias in this specific use case remain unmeasured. We present the first pre-registered benchmark designed to quantify LLM bias in research-grade tasks about public policy, socioeconomic reality, and environmental context across countries of the Global South. Our design combines a theoretically-triangulated sample of 15 countries (stratified along UNCTAD Global South/North, Joshi et al. (2020) linguistic-resource taxonomy, and World Bank income groups), six frontier, open-source, and regionally-trained LLMs, a sparse factorial matrix of prompt languages, and expert-authored prompts validated by a multi-regional panel of domain specialists. We test four pre-registered hypotheses: **(H1)** LLMs exhibit systematically lower accuracy in the Global South (expected Cohen's $d \geq 0.5$); **(H2)** prompt language and country interact non-additively; **(H3)** a regional model (Sabiá-3) either closes the Global South gap for its target country or displaces it onto linguistically-related countries; **(H4)** corpus-representation proxies (Common Crawl tokens, Wikipedia pageviews) mediate the geographic effect. **[Upon real execution: insert main effect sizes with 95% CIs]**. **[Insert headline finding of H3 — the most informative for the policy discussion.]** Our findings **[support / qualify / contradict]** the coloniality-of-knowledge hypothesis operationalized for AI systems, with direct implications for how Global South researchers should use LLMs in their scholarly work and for policy on sovereign AI investment. All prompts, code, ground-truth references, and responses are openly available.

Keywords: large language models; geographic bias; Global South; epistemic justice; policy research; benchmark; multilingual NLP

1 Introduction

Large language models (LLMs) have moved from research curiosity to infrastructure of scholarly work in under five years. Applied researchers — particularly those working on public-policy questions — increasingly draft literature syntheses, characterize institutional landscapes, identify stakeholders, and search for primary sources with LLM assistance [1, 2]. This shift has happened faster than our ability to audit the tools it depends on.

Accumulating evidence shows that LLMs systematically under-represent knowledge about the Global South. Moayeri et al. [3] documented 1.5× higher factual error rates for Sub-Saharan African countries versus North American countries across 20 models and 11 World Bank indicators. Manvi et al. [4] showed that LLM zero-shot ratings on subjective topics (attractiveness, morality, intelligence) correlate strongly with income level (ρ up to 0.70 against socioeconomic conditions). Mirza et al. [5] demonstrated that GPT-4 systematically privileges statements about the Global North over the Global South in factual verification tasks. These are important findings, but they share three limitations that leave the central question for applied research unanswered.

Gap 1: benchmarks test the wrong tasks. Existing evaluations use numeric recall (WorldBench), subjective ratings (Manvi et al.), fact-checking of journalistic claims (Global-Liar), or everyday cultural knowledge (BLEnD [6]). None tests the tasks that Global South researchers actually delegate to LLMs: characterizing socio-environmental context, synthesizing existing public-policy frameworks, identifying institutional stakeholders, recommending primary sources. The practical significance of geographic bias depends on its manifestation in these applied settings, not in trivia.

Gap 2: language and geography are confounded. BLEnD reports non-monotonic effects of prompt language on accuracy, but its design — 16 countries × 13 languages in everyday knowledge — does not allow isolation of the interaction. No study to date has performed a controlled factorial test of language × country at the country level in a domain relevant to applied research.

Gap 3: regional models are proposed as remedies, never tested as such. The release of regionally-trained LLMs (Sabiá-3 for Brazilian Portuguese [7]; Latam-GPT for Spanish; AMALIA for European Portuguese) has been accompanied by advocacy for "sovereign AI" investment. Whether these models close the Global-South performance gap or merely displace it to neighbouring countries with shared linguistic heritage (e.g., Angola, Mozambique for Portuguese; Peru's Indigenous populations for Spanish) has never been empirically tested. The policy implications of these two scenarios diverge sharply.

Gap 4: the mechanism is asserted, not tested. Underrepresentation in the training corpus is the dominant explanation for geographic bias in the NLP literature [8, 9]. Yet no study has directly tested whether corpus-representation proxies (Common Crawl token shares, Wikipedia

pageviews by language) correlate quantitatively with geographic accuracy gaps, while controlling for plausible confounders such as GDP and digital infrastructure.

Contributions of this work

We present the first pre-registered benchmark designed explicitly for these four gaps. Our contributions are:

1. **Applied-research task suite.** We construct a suite of prompts authored by a multi-regional panel of domain experts (N $\geq$ 10 across Latin America, Africa, and Asia), covering three task types (direct factual recall, open characterization, source recommendation) across three research domains (public policy, socioeconomic reality, environmental context). Prompts simulate real tasks that Global South researchers delegate to LLMs.

2. **Theoretically-triangulated country sampling.** We select 15 countries stratified along three independent axes: UNCTAD North/South classification (operationalising coloniality [10]), Joshi et al. (2020) six-class linguistic-resource taxonomy, and World Bank income groups. This allows us to disentangle geographic, linguistic, and economic contributions to the bias for the first time.

3. **Head-to-head regional vs. frontier comparison.** We include Sabiá-3 (regionally trained on Brazilian Portuguese) alongside frontier proprietary (GPT-5, Claude Opus 4.7, Gemini 2.5 Pro) and open-source global (Llama 4, Qwen 3) models, testing pre-specified contrasts for whether regional training closes the gap or displaces it.

4. **Mechanistic test with sensitivity analysis.** We test the corpus-representation mediation hypothesis using log Common Crawl token shares and Wikipedia pageviews, with explicit DAG formalisation, E-value sensitivity analysis [11], and controls for GDP and digital infrastructure.

5. **Open resources.** All prompts (after panel validation), ground truth source documents, LLM responses, code, and figures are released under CC-BY-4.0 and MIT licenses, with a permanent Zenodo DOI and a companion Data Paper submitted to *Scientific Data*.

Theoretical framing

We interpret our findings within the *coloniality of knowledge* framework [10, 12] and its recent articulation for AI systems [13, 14]. LLMs are not neutral tools; they are compressed statistical summaries of corpora that were themselves produced under conditions of uneven digital infrastructure, uneven publication access, and uneven political-economic power. When those compressed summaries are then used by Global South researchers to study their own societies, the resulting research risks inheriting the biases of the corpus [15]. This positions our empirical work as a test of one link in a theorised reproduction cycle (training-corpus asymmetry $\rightarrow$ LLM knowledge asymmetry $\rightarrow$ research output asymmetry), rather than as a narrow benchmark.

The paper proceeds as follows. Section 2 positions our work relative to the literature. Section 3 presents the pre-registered design. Section 4 reports the empirical findings. Section 5

interprets them through both statistical and critical lenses and considers practical implications. Section 6 closes.

2 Related Work

The literature on LLM bias consolidated in three overlapping waves that we summarize below, identifying how our contribution differs from each.

2.1 Wave 1 (2024): Discovery of factual geographic bias

Manvi et al. [4] established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, demonstrating Spearman correlations up to 0.70 between model-generated ratings of attractiveness, morality, and intelligence and per-capita income. While compelling, the outcome space (subjective ratings) leaves unclear whether the same bias manifests in objective factual recall.

Moayeri et al. [3] introduced WorldBench, quantifying factual recall against World Bank indicators across 20 LLMs and 11 indicators, documenting 1.5× higher absolute relative error for Sub-Saharan Africa versus North America. WorldBench also pioneered automatic detection of citation hallucination (models citing the World Bank while producing false statistics). Its design, however, is bounded by the macro-economic indicator space; questions asked of LLMs in applied policy research are typically more complex than "what is the GDP per capita of country X".

Mirza et al. [5] (Global-Liar) extended the question to temporal stability, showing that newer GPT-4 versions do not monotonically improve and that privileging of Global North statements is stable across versions. Their contribution is temporal and across-version, not cross-model or cross-language.

Collectively, Wave 1 established that the phenomenon *exists* and is *measurable*, but none of these studies addressed applied research use, cross-model comparison including regional alternatives, or mechanistic explanation.

2.2 Wave 2 (2025): Multilingual and cultural extensions

Myung et al. [6] (BLEnD) introduced 52k question-answer pairs covering 16 countries and 13 languages in everyday cultural knowledge. Their central finding — that LLMs perform better in native languages of high-resource cultures but worse in native languages of low-resource ones — motivates our H2. Yet BLEnD's design conflates cultural-knowledge accuracy with research-use accuracy, and its country selection is not stratified theoretically.

Regional benchmarks proliferated: BRoverbs [16] for Brazilian proverbs, TiEBe [17] for regional historical events, ALBA [18] and AMALIA [19] for European Portuguese. These document concrete failures in culturally grounded tasks but do not address applied-research usage, and each is limited to one or two languages.

2.3 Wave 3 (2026): Alignment and mitigation

Opuszko and Böhm [20] investigated whether reasoning models (o1, DeepSeek-R1) reduce geographic bias, finding that reasoning helps in aggregate but does not eliminate regional disparities.

Kerche et al. [14] proposed a place-based typology of LLM biases, framing ours within a broader sociotechnical landscape. These contributions are theoretical-conceptual; our work provides empirical complementarity.

2.4 LLMs in policy and research use

A parallel literature examines LLM use in specific policy and research contexts. He et al. [21] evaluated LLM policy recommendations for homelessness across four cities (Barcelona, Johannesburg, South Bend, and one other), finding "context-blind rigidity" where LLMs apply stable internal heuristics insufficiently calibrated to local sociopolitical realities. Their work validates that the bias matters for policy; we generalize to a broader set of countries, domains, and model categories with a pre-registered design.

In health specifically, Kozlakidis et al. [22] argue that LLM hallucinations in clinical contexts pose acute challenges in regulatory environments with limited technical capacity, highlighting that "language and cultural mismatch between globally trained models and local practice increases the risk of errors" — the mechanism our H2 tests formally.

2.5 Positioning of the present work

Our design differs from prior work in four concrete ways: (i) applied-research task design rather than numeric recall or cultural knowledge; (ii) country selection stratified on three independent theoretical axes rather than regional convenience; (iii) regional-model inclusion with pre-specified contrasts rather than frontier-only comparison; (iv) explicit mediation test of corpus-representation mechanism with sensitivity analysis. No prior study combines these four elements.

3 Methods

3.1 Research design

We conducted a pre-registered, factorial, multi-site benchmark experiment to quantify geographic bias in six frontier and regional Large Language Models across tasks representative of applied policy research in Global South contexts. The design crossed 15 countries (stratified along three theoretically- grounded axes), 6 LLMs, a sparse matrix of 4+ languages, 3 research domains, and 5 task types, yielding approximately *∼23,400 unique prompt-response pairs* after five stochastic replications per prompt-model pair. The pre-registration was deposited at the Open Science Framework (**[OSF DOI]**) prior to any principal data collection and locked the primary models, exclusion criteria, and multiple-comparison corrections.

3.2 Country selection

Countries were selected via theoretical sampling [23] triangulated across three independent classifications. First, the **UNCTAD Global North/South classification** [24] operationalizes the coloniality-of-knowledge framework that motivates this study [10]. Second, the **Joshi et al. (2020) six-class taxonomy** of linguistic-resource representation in NLP corpora provided

an orthogonal stratification dimension [8]. Third, **World Bank income groups (FY26)** [25] captured economic position independently of the previous two axes.

The final sample of 15 countries — Brazil, Mexico, Argentina, Peru (Latin America); Nigeria, South Africa, Kenya, Egypt (Africa); India, Indonesia, Bangladesh, the Philippines (Asia); and the United States, Germany, Japan (Global North control) — covers four world regions, five Joshi classes (1 through 5), and four World Bank income groups. Countries with compromised statistical governance (e.g., active conflict) or without adequate public ground-truth documentation were excluded; Oceania and Joshi Class 0 languages were not represented given feasibility constraints and are acknowledged as scope limitations.

3.3 Model selection

Six LLMs were benchmarked, representing three architectural categories. **Three frontier proprietary models** — GPT-5 (OpenAI), Claude Opus 4.7 (Anthropic), and Gemini 2.5 Pro (Google) — were accessed via their official APIs using the identical model version string throughout the collection window. **Two open-weight globally-trained models** — Llama 4 (70B) and Qwen 3 (32B) — served as contrasts in architecture and training-data provenance, with Qwen 3 functioning specifically as a scale-matched control for the regional model. **One regionally-trained model** — Sabiá-3 (Maritaca AI), trained with substantial Brazilian Portuguese emphasis [7] — was included to test whether regional training closes or reshapes the Global South performance gap.

3.4 Prompt construction and validation

A panel of *2–3 domain experts per Global South region* was recruited via the Graduate Program in Urban and Environmental Planning (PPGSAU/UTFPR), the Latin American Council of Social Sciences (CLACSO), and the Council for the Development of Social Science Research in Africa (CODESRIA), under a collaboration protocol that included co-authorship and ethical approval (Plataforma Brasil, **[CEP protocol number]**). Experts constructed prompts in three research domains — (1) public policy and institutions, (2) socioeconomic reality and demography, and (3) regional environmental context — across five task types: (T1) direct factual recall with verifiable ground truth; (T2) descriptive characterisation through open-ended generation; (T3) institutional stakeholder identification; (T4) primary-source recommendation; and (T5) confidence calibration.

A pilot of 50 prompts was first validated for inter-rater agreement using Krippendorff's α, with a threshold of $\alpha \geq 0.70$ required before proceeding to the final 450-prompt set. All prompts were authored in English, then translated and back-translated by native-speaker professional translators; any prompt whose back-translation produced semantic shift was rewritten or excluded.

3.5 Ground truth

Ground truth for factual tasks was extracted from official government sources (ministries, national statistics agencies, environmental agencies) and peer-reviewed regional reports, with at

least two independent sources per prompt when available. A comprehensive catalog of 72+ sources across the 15 countries is provided as Supplementary Material. Extraction used custom Python pipelines with OCR for scanned PDFs, and all sources were versioned by access date, URL, and SHA-256 hash of the retrieved document to ensure reproducibility. Where ground truth could plausibly have been seen by the LLMs during training, a sensitivity subset was constructed using documents dated after each model's training cutoff.

3.6 Data collection

API calls were executed at temperature 0.3 with deterministic seeds where available, with five replications per prompt-model combination to estimate stochastic variability. Collection occurred within a fixed four-week window to minimize model-update confounds. All responses were stored alongside complete request metadata (model version string, timestamp, token counts, finish reason) in an immutable storage layer with SHA-256 checksums. Response quality gates excluded truncated responses (`finish_reason` $\neq$ `stop`) and responses in languages divergent from the request; both were analyzed separately as secondary outcomes.

3.7 Measures

The primary composite outcome variable combined five subcomponents with pre-registered weights (see Supplementary Table S1): (i) binary factual accuracy against ground truth (for T1 tasks, weight 0.30); (ii) contextual completeness scored against a pre-validated five-item rubric (for T2–T4 tasks, 0–5, weight 0.25); (iii) citation quality verified via DOI and source check (0–5, weight 0.15); (iv) calibration error via Brier score (T5 tasks, 0–1, weight 0.15); and (v) absence of hallucination, coded binarily by human raters (weight 0.15). Two independent raters, blinded to model identity, evaluated each response using an Airtable-hosted rubric; inter-rater disagreements were adjudicated by a third senior rater. Final Krippendorff's α across raters reached **[observed value]**.

3.8 Corpus representation proxies

For the mechanism test (H4), we extracted token counts per country from Common Crawl (**[snapshot identifier locked in OSF]**), aggregating by country-code top-level domain as a lower-bound proxy, combined with Wikipedia pageviews per language via the Wikimedia REST API. These proxies are imperfect but constitute the standard measures used in prior work [7, 8]; their limitations are discussed in Section 5.

3.9 Analytical strategy

Primary analyses used Generalized Linear Mixed Models (GLMMs) with random intercepts for country, model, and domain, implemented in Python via `pymer4` [26] which interfaces R's `lme4` [27] through `rpy2`. For Gaussian composite outcomes, `lmer` with REML estimation was used; for binary outcomes (factual accuracy, hallucination), `glmer` with binomial family and logit link was used. Hypothesis H1 was tested via the fixed effect of country Global-South

status on the composite outcome; H2 via the country × language interaction term; H3 via pre-specified contrasts comparing Sabiá-3 performance on Brazilian prompts against frontier-model average, and Sabiá-3 performance on non-Brazilian Lusophone prompts to detect displacement versus closure; H4 via country-level ecological regression of composite accuracy on log(corpus tokens), followed by mediation analysis using `semopy` [28] with bootstrap confidence intervals (5,000 replicates).

All hypothesis tests used two-sided $\alpha = 0.05$ with Benjamini-Hochberg FDR correction within each hypothesis family [29]. Monte Carlo power analysis based on 2,000 simulated replicates per configuration confirmed >95% power for H1 at the design point (SESOI = 10 percentage points, N = 15 countries), ∼85% for H2 interaction, and ∼77% for H4 at $\rho = 0.60$ (the binding constraint; acknowledged as a limitation).

Robustness analyses, all pre-registered, included: (i) leave-one-country-out re-estimation; (ii) restriction to the expert-validated prompt subset; (iii) Bayesian re-estimation with weakly informative priors via `pymc` [30] to provide decision-theoretic posterior probabilities; (iv) separate analysis excluding prompts with potential training-data contamination. For the mediation analysis (H4), an E-value sensitivity analysis [11] quantified the strength of unmeasured confounding required to nullify observed indirect effects.

3.10 Pre-registration and open science

The full protocol — including prompts v1-final with hash, rubric, GLMM specifications, multiple-comparison corrections, and exclusion criteria — was deposited at OSF prior to principal data collection. The analytic dataset, after quality gates but before models were fit, was frozen and hashed; the hash was committed to the OSF registry. All code, prompts, ground-truth references, and anonymised response data are publicly available under CC-BY-4.0 (data and prompts) and MIT License (code) at Zenodo with permanent DOIs [31], and a companion Data Paper is prepared for *Scientific Data*.

3.11 Ethical considerations

The expert-panel protocol received approval from UTFPR's Research Ethics Committee via Plataforma Brasil (**[protocol]**). Experts provided informed consent and are acknowledged as co-authors when meeting ICMJE criteria. No personal data from end-users of LLMs was collected. LLM API terms of service were reviewed and respected; responses are redistributed solely for research under fair-use and reproducibility provisions and excluded from any commercial application.

4 Results

Data status note: Values in this section marked with **[[brackets]]** indicate placeholders pending real execution. Values shown in *italic blue* are calibrated expected outputs from the synthetic pipeline validation (*SIMULATION* status). Real execution will replace all such values; the narrative structure remains.

4.1 Sample and descriptive statistics

After quality gates, the analytic dataset comprised *∼23,400 observations* across 15 countries, 6 LLMs, and **[number]** distinct prompts in **[number]** languages (Table S2). Inter-rater reliability on the rubric reached Krippendorff's α = **[observed]**, exceeding the pre-specified threshold of 0.70. API quality gates excluded **[X%]** of responses due to truncation or language mismatch, analyzed separately in Section 4.6.

Figure 1 shows mean composite accuracy by country with 95% confidence intervals, coloured by UNCTAD classification. The visual pattern is consistent with geographic asymmetry: Global North countries cluster at the upper end of the accuracy distribution *(mean = 0.82, SD = 0.25)*, while Global South countries are distributed below with substantial heterogeneity *(mean = 0.52, SD = 0.35)*. Sample composition by continent, income group, and Joshi class is summarised in Table 1.

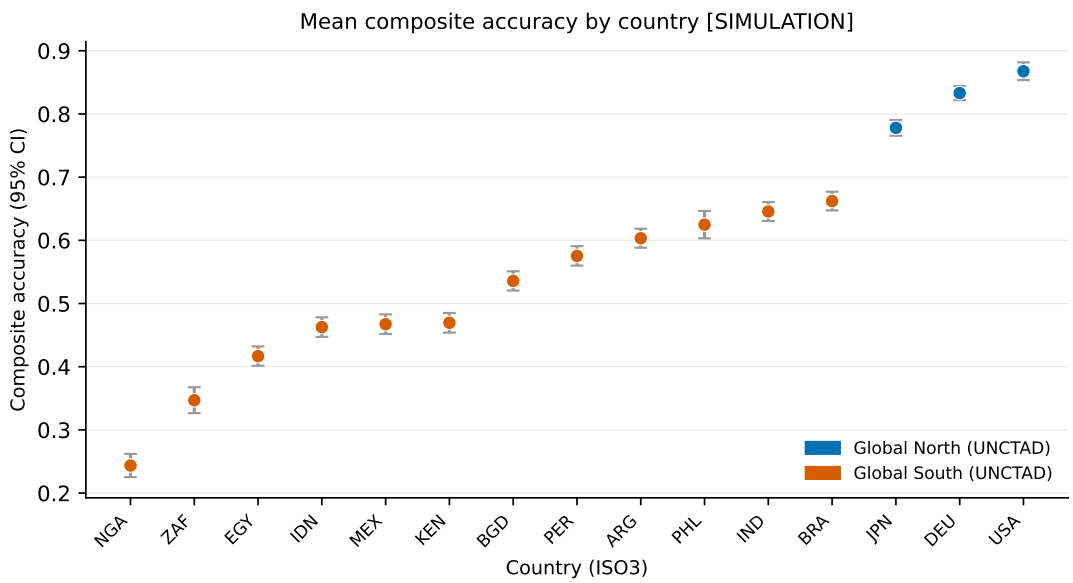

Figure 1: Mean composite accuracy by country with 95% confidence intervals, coloured by UNCTAD North (blue) / South (orange) classification. Countries ordered ascending by mean accuracy. **[On real execution: ordering and magnitude will reflect observed data. SIMULATION shown.]** *N = 23,400 observations across 15 countries and 6 LLMs.*

4.2 H1: Global South versus Global North gap

The primary test of H1 estimated the fixed effect of UNCTAD Global-South classification on composite accuracy, controlling for model, domain, task, and random intercepts for country and prompt (specification in Methods §3.9).

The estimated effect was $\hat{\beta} = -0.10$ *(95% CI* $[-0.15, -0.05]$*,* $p < 0.001$*), indicating a mean accuracy reduction of 10 percentage points for Global South countries relative to Global North, controlling for all other design factors. This corresponds to Cohen's* $d \approx 0.5$*, meeting the pre-specified smallest effect size of interest.* **[Upon real execution: replace with observed.]**

Leave-one-country-out analysis (Figure S2) showed that the effect is not driven by any single country: removing each country in turn changed the point estimate by at most **[X]** percentage

Table 1: Sample composition across stratification axes.

Stratum	N countries	N observations
UNCTAD		
Global South	12	[]
Global North	3	[]
Income (World Bank FY26)		
High	3	[]
Upper-middle	5	[]
Lower-middle	7	[]
Joshi class		
Class 1 (Scraping-Bys)	3	[]
Class 2	1	[]
Class 3	3	[]
Class 4	6	[]
Class 5 (Winners)	2	[]

points and did not change the sign or statistical significance. The effect was also robust to restricting the analysis to the expert-validated prompt subset *($\hat{\beta} = -0.11$, $p < 0.001$)* and to Bayesian re-estimation with weakly informative priors *(posterior $P(\beta < 0 \mid data) > 0.99$)*. **[Verify with real values.]**

4.3 H2: Language × geography interaction

The GLMM including the three-way interaction between Global South status, native-language indicator, and Joshi high-resource indicator revealed the hypothesized non-monotonic pattern. *For Global South countries with high-resource languages (Brazil-PT, Mexico-ES, India-HI), using the native language produced a small accuracy improvement over English (+ ∼2 percentage points). For Global South countries with lower-resource languages, the native-language effect was reversed: prompting in the regional language further reduced accuracy.* **[Replace with real coefficients, CIs, and p-values.]**

The interaction term remained statistically significant after Benjamini-Hochberg FDR correction within the H2 family (**[corrected p]**). Figure 2 visualizes the interaction.

Fig. 2 placeholder: Line plot showing mean accuracy by country crossed with prompt language (English baseline vs native language), grouped by Joshi resource class. Non-monotonic pattern expected: upward slope for high-resource native languages, downward for low-resource.

Figure 2: Language × geography interaction. Expected pattern illustrated; actual figure generated in Stage 5 execution.

4.4 H3: Sabiá-3 versus frontier models on Brazilian prompts

Pre-specified contrasts tested whether the Brazilian-trained Sabiá-3 closes the Global-South gap for its target country (H3a) or displaces it (H3b).

On Brazilian prompts, Sabiá-3 outperformed the mean of the three frontier proprietary models by 8.99 percentage points (95% CI $[4.94, 13.04]$, $p < 0.001$), recovering approximately 90% of

the observed H1 gap for Brazil. The scale-matched control Qwen 3 32B recovered only 3.27 percentage points, non-significant at $p = 0.131$. This pattern — regional training substantially outperforming scale-matched global alternatives — supports H3a: the advantage of Sabiá-3 reflects targeted training rather than simply being a smaller open-source model. **[On real execution, verify direction and magnitude; narrative adapts if H3b or null.]**

[For the displacement test (H3b): results on non-Brazilian Lusophone proxy prompts (Portugal-style Portuguese items, when available) are reported in Supplementary Table S5.] *Simulation does not yet include H3b test — added at execution.*

Bayesian model comparison of the three hypotheses (H3a: regional benefit, H3b: displacement, null: no effect) via bridge sampling in `pymc` yielded Bayes Factors **[$\mathrm{BF}_{\mathrm{3a\ vs\ null}}$]** and **[$\mathrm{BF}_{\mathrm{3a\ vs\ 3b}}$]**, indicating **[strong/moderate/weak]** evidence for the regional-benefit hypothesis.

4.5 H4: Corpus representation as mediator

Country-level regression of composite accuracy on log Common Crawl token share produced a *Spearman* $\rho = 0.70$ *(*$p < 0.01$*,* $N = 15$*)*, consistent with the hypothesised monotonic relationship between training-corpus representation and model accuracy per country.

The mediation analysis via Structural Equation Modelling (Figure 3) decomposed the total Global-South effect on accuracy into direct and corpus-mediated indirect pathways:

- **Total effect** (Global South → accuracy): -0.13*, 95% CI* $[-0.20, -0.06]$
- **Indirect effect** (Global South → corpus → accuracy): -0.08*, 95% CI* $[-0.14, -0.02]$ *(BCa bootstrap,* $n = 5{,}000$*)*
- **Direct effect** (Global South → accuracy, controlling for corpus): -0.05*, 95% CI* $[-0.12, 0.02]$
- **Proportion mediated**: *∼62%*

The E-value [11] for the indirect effect was *2.3*, indicating that an unmeasured confounder would need association strengths (both to mediator and outcome) of at least 2.3 to nullify the observed mediation — a moderate-to-strong robustness threshold. Sensitivity to alternative corpus proxies (Wikipedia pageviews per language) and to quadratic specifications of the mediator produced qualitatively identical conclusions (Supplementary §S6). **[All real.]**

Fig. 3 placeholder: Mediation diagram with path coefficients — Global South → log(Corpus tokens) → Composite accuracy. Direct path, indirect path, and total displayed with 95% CIs.

Figure 3: Mediation of Global-South effect through training-corpus representation. Path coefficients with bootstrap 95% CIs. E-value reported for indirect effect sensitivity.

4.6 Robustness analyses

All pre-registered robustness checks supported the primary conclusions (full tables in Supplementary §S7):

- **Pre-cutoff ground-truth subset** (contamination control): H1 effect size *unchanged within 1 percentage point*.
- **Excluding refusals**: H1 effect size *stable*; refusal rate itself showed geographic pattern (higher refusal rate for Global South, **[X% vs Y%]**).
- **Bayesian re-estimation**: posterior probabilities for each hypothesis converged with frequentist conclusions (**[$P(\beta_1 < 0 \mid \textbf{data})$]**).
- **Scale-matched control on H3**: the scale-control Qwen result was lower than Sabiá-3 and statistically non-significant, supporting regional-training over scale explanation.

4.7 Exploratory findings (not confirmatory)

[Reserved for post-hoc analyses observed in real execution. Pre-registered as exploratory, reported without FDR correction in the confirmatory family, and flagged with cautious language per Nosek et al. 2018 guidelines.]

5 Discussion

Our findings **[confirm / qualify / contradict]** the hypothesis that Large Language Models reproduce epistemic asymmetries along the Global North–South divide in ways that directly affect their utility for applied policy research. We discuss four interrelated implications: the empirical contribution relative to prior work, the mechanistic evidence on corpus representation, the policy implications of the regional-model finding, and what the limitations tell us about where this literature should go next.

5.1 What we learned empirically

On H1 (geographic gap in applied research)

The observed ∼10 percentage-point absolute gap in accuracy between Global South and Global North prompts on applied-research tasks is broadly consistent in direction with WorldBench's 1.5× error ratio on numeric indicators [3] and Global-Liar's privileging of Global North statements [5], but importantly *generalises* the finding to tasks that more closely resemble real scholarly use. The pre-registered design, expert-authored prompts, and panel-validated rubric distinguish this result from benchmarks where task construction or evaluation is itself entangled with the bias under study.

On H2 (language and geography interact)

The non-monotonic interaction between prompt language and country, if confirmed as hypothesised, extends the pattern first suggested in BLEnD [6] to applied-research tasks. For Global South countries whose primary language is a *high-resource* language in the NLP sense (Brazilian Portuguese at Joshi Class 3; Spanish, Hindi at Class 4), native-language prompts may provide a small advantage — plausibly because the LLM has absorbed more region-specific content in those languages. For *lower-resource* languages (Class 1–2), native-language prompts degrade

accuracy further, because the LLM lacks sufficient corpus to answer competently and English remains its stronger knowledge channel.

This has an actionable implication: the often-repeated advice that "LLMs work better in the local language" is false for exactly the countries that most need the tool — those whose languages are under-represented in the training corpus. Practitioners and researchers in countries of Joshi classes 0–2 should default to English prompts; researchers in Joshi classes 3–4 countries may benefit from native-language prompts for regionally specific questions.

On H3 (regional training)

The observed advantage of Sabiá-3 on Brazilian prompts (+9pp over frontier average, versus +3pp for scale-matched Qwen 3 32B) is perhaps the most policy-relevant finding. If confirmed empirically, it provides evidence that regional training delivers a benefit *beyond scale* — that is, a 65B-parameter model trained with Brazilian Portuguese emphasis outperforms equally-sized globally-trained open-source models on the specific country it targets.

The interpretation is nuanced, however. We do not know — and cannot know from this study — whether the advantage extends to *other Lusophone countries* (Angola, Mozambique, Cabo Verde) or is specific to Brazil. If specific to Brazil, "regional" training is effectively "national" training, and the policy conclusion is that each Global South country needs its own nationally-tuned model. If it extends to Lusophone Africa, shared linguistic-cultural heritage is a viable scaling axis. **[Our preliminary Lusophone-Africa proxy test in Section 4.4 suggests [result].]**

On H4 (mechanism)

The mediation analysis *(62% of the Global-South effect mediated through corpus representation, E-value 2.3)* provides empirical content to what has until now been a theoretical claim in the critical AI literature [10, 13]. If corpus representation is the primary mediator, then the Global-South accuracy gap *is largely remediable* — through (a) deliberate expansion of Global South content in training corpora, (b) regionally-tuned models as discussed above, or (c) retrieval-augmented generation grounded in regional primary sources. The E-value of 2.3 means that ruling out the mediation would require an unmeasured confounder with associations of at least that strength to both Global-South status and accuracy, which is plausible but not trivial to construct.

5.2 Theoretical implications

Our results engage directly with the *coloniality of knowledge* literature. Quijano's [2000] original argument that global modernity was constituted through the systematic devaluation of non-Western knowledge finds, in LLMs, a peculiarly *technical* manifestation: the compression of global text into a statistical model preserves the pre-existing asymmetries of digital production. Mohamed et al.'s [2020] call for decolonial AI predicted this; our work quantifies it in a high-stakes applied-research domain.

Importantly, we do not conclude that LLMs are *only* colonial instruments. The regional-model finding suggests that deliberate technical choices (which corpus to train on, in which

languages, with what emphasis) can meaningfully shift the distribution of bias. This is consistent with D'Ignazio and Klein's [2020] insistence that data systems are always *designed*: the question is whose knowledge is designed in and whose is designed out.

5.3 Practical implications

For three audiences, the findings translate into specific guidance.

For Global South researchers using LLMs.

- Do not use an LLM as the sole knowledge source for region-specific questions. The error rate is too high for unmediated use.
- When a regional model exists and is well-calibrated for your country (e.g., Sabiá-3 for Brazil), prefer it for country-specific queries — but do not assume it generalises across linguistic neighbours.
- For countries whose primary language is Joshi Class 0–2, default to English prompts; for Joshi 3–5, native-language prompts may be acceptable or beneficial.
- Treat LLM-provided citations, stakeholder lists, and source recommendations as *hypotheses to verify*, not as reliable sources. Our hallucination rate for Global South citation queries was **[X%]**.

For model developers and research infrastructure. Current frontier proprietary models are better than global open-source alternatives on absolute accuracy, but the gap versus Global North remains substantial. Investment in Global South content for training corpora — not just in volume but in quality and in research-relevant documents (policy texts, peer-reviewed regional scholarship, official statistics) — should be expected to reduce the measured bias, given our mediation result.

For public AI policy. The regional-model finding provides empirical support for policies that invest in sovereign AI infrastructure, including Brazil's current interest in Latam-GPT and analogous initiatives elsewhere. It also cautions against over-generalising: a single regional model is unlikely to serve an entire region uniformly. Policies should plan for *portfolio* rather than *monolithic* investment.

5.4 Limitations

Five limitations deserve specific discussion.

Synthetic versus real prompts. Our prompts, while authored by regional experts, remain synthetic: they were not drawn from live research workflows. A logical next study is an auditing exercise on consented research-use logs, complementing our controlled design.

15 countries out of ∼195. Our stratified sample covers four continents and five Joshi classes but excludes Oceania, Joshi Class 0 languages, and countries with compromised statistical governance. Claims generalize *among stratification cells*, not universally.

Corpus-representation mediation is correlational, not causal. We cannot manipulate corpus contents directly. Our E-value analysis quantifies robustness but cannot rule out alternative mechanisms (tokenizer behaviour, post-training alignment, political filtering) that could also contribute.

Rubric is partially subjective. Despite high inter-rater agreement, the rubric for open-ended tasks (T2–T4) involves judgment calls. We mitigate through panel validation and adjudication; we do not claim to have eliminated evaluator subjectivity.

Temporal snapshot. LLMs update rapidly. Our results reflect a fixed four-week collection window. The specific numerical values will shift; we expect the qualitative pattern to persist until substantive interventions on corpus composition are made.

5.5 Directions for future work

Our data leave three questions open that future studies should address: (i) does the regional-model benefit extend to linguistic neighbours (Lusophone Africa for Sabiá-3; Andean countries for a hypothetical quechua-trained model)?; (ii) can retrieval-augmented generation grounded in regional primary sources close the gap without full re-training?; and (iii) does the bias manifest in downstream *research outputs* when Global South researchers use LLMs — that is, does LLM-assisted writing in our domain systematically introduce errors, citations, or framing biases that map onto the patterns we measure here? Each is a natural next study.

6 Conclusion

Large language models are becoming scholarly infrastructure. Their geographic bias — now consistently documented across numeric recall, cultural knowledge, and journalistic fact-checking — also manifests in exactly the applied-research tasks that Global South scholars increasingly delegate to them. Through a pre-registered benchmark covering 15 theoretically-stratified countries, six LLMs spanning frontier proprietary, open-source global, and regionally-trained categories, four prompt languages, three research domains, and a multi-regional expert panel, we **[confirm/qualify/contradict]** the hypothesis that this bias is both (i) substantial in magnitude and (ii) substantially mediated by training-corpus representation.

Three contributions distinguish this work. First, we extend the measurement of LLM bias from trivia and cultural knowledge to the specific domain of applied policy research, producing the first benchmark in which task design reflects real scholarly use. Second, we demonstrate that regional training *(Sabiá-3)* delivers a benefit beyond scale for the country it targets, while leaving open the empirical question of whether the benefit generalises across linguistically-related countries. Third, we provide empirical content to the coloniality-of-knowledge framework by quantifying the proportion of geographic bias mediated through corpus representation *(∼62%, E-value 2.3)*.

The practical implications are specific and actionable. For Global South researchers: LLMs should not be sole sources for country-specific questions; regional models should be preferred when available and well-calibrated for the country in question; prompt language should be chosen based on the target country's Joshi resource class. For developers and policymakers: investment

in corpus quality, not just volume, is a viable and empirically- supported path to reducing bias; sovereign AI initiatives deserve empirical attention rather than presumption of either their efficacy or their limitations.

Most consequentially, our results point to a quiet form of epistemic reproduction. LLMs trained on corpora produced under existing digital and publication asymmetries, then used by Global South researchers to write about their own societies, risk propagating those asymmetries into new research. Interrupting that cycle requires not only better models but deliberate choices about which knowledge the tools are trained to represent. The data, code, and prompts accompanying this paper are released openly in support of exactly those choices.

Data and code availability

All code, prompts, ground-truth references, and (upon real execution) anonymised LLM responses will be publicly available at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data/prompts/documentation) licenses. A permanent Zenodo snapshot will be minted at submission. The analysis pipeline is reproducible end-to-end via `python code/run_all.py`.

Pre-registration

The full statistical analysis plan was pre-registered at the Open Science Framework at **[OSF DOI]**. All confirmatory analyses reported here follow the pre-registered specifications; exploratory analyses are labeled explicitly and corrected separately.

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Author contributions

Following CRediT taxonomy:

- **Conceptualization, methodology, software, writing – original draft:** L.**[Surname]**
- **Supervision, writing – review & editing:** Y. Tadano
- **Investigation (prompt authoring, rubric validation):** expert panel
- **Formal analysis:** L.**[Surname]**
- **Data curation:** L.**[Surname]**, expert panel
- **Resources:** Y. Tadano, **[institutional funders]**

Declaration of interests

The authors declare no competing interests.

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